

**5.22** Solution: The result can be obtained by simple arguments using the results from Problems 5.17, 5.19, and 5.20.

The magnetic field from a bar with uniform magnetization can be obtained from either magnetic scalar potential or from vector potential. If we use the vector potential,  $\nabla \times \mathbf{M} = 0$ , and the effect of the magnetization can be captured by surface currents  $\mathbf{M} \times \mathbf{n}$  around the cylinder, on the curved surface. When this bar is placed against an infinitely permeable flat surface, the surface currents will induce image currents inside the medium, with the same magnitude and direction, from Problem 5.17. These image currents are equivalent to an image bar with the same magnetization as the original bar, placed against the flat surface from inside the medium, with its top surface in full contact with the bottom surface of the original bar. From Problem 5.19, the magnetic induction at the bottom surface of the original bar can be approximated as

$$\mathbf{B} \approx \frac{\mu_0 M}{2} \frac{L}{\sqrt{a^2 + L^2}} \hat{z} \approx \frac{\mu_0 M}{2} \hat{z}.$$

Here,  $a$  can be viewed as the horizontal length scale of the bar. If  $a \ll L$ , the magnetic induction on the bottom surface can be taken as constant and is the value on the axis. Since the image bar has a uniform magnetization  $M\hat{z}$ , applying the result from Problem 5.20, we have

$$F = \oint (\mathbf{M} \cdot \mathbf{n}) \mathbf{B} da = M \cdot \frac{\mu_0 M}{2} \hat{z} \cdot A = \frac{\mu_0}{2} A M^2 \hat{z},$$

which is a force in the  $+\hat{z}$ -direction, pulling the image bar to the original bar, *i.e.*, an attractive force.